

Study Notes on “Agency MBS as Safe Assets” (He & Song, RFS 2026)

Notes (models, tables, and figures)

Contents

1	What the paper is trying to do	3
2	Theoretical framework: demand for safe assets, pricing, and issuance	3
2.1	Preferences and the safe-asset “liquidity” aggregator	3
2.2	Budget constraint and Euler equations	4
2.3	Convenience premiums: one-period yields and a log approximation	4
2.4	Supply side: issuance by MBS intermediaries	5
2.5	Equilibrium system	5
3	Comparative statics: propositions and proofs (fully spelled out)	5
3.1	Proposition 1: changes in MBS-specific safety/liquidity λ_t^M	5
3.2	Proposition 2: changes in overall safe-asset demand γ_t	6
3.3	A crowding-out implication (Treasury supply)	7
4	From theory to empirics: how the model objects are measured	8
4.1	Key observables and their model interpretation	8
4.2	Prepayment-driven demand: the economic mechanism and empirical proxy	8
5	Figures and tables: what each one shows and how I interpret it	9
5.1	Figure 1: HQLA and repo (why agency MBS matter as “safe assets”)	9
5.2	Figure 2 and Table 1: measuring and quantifying the MBS convenience premium	9
5.3	Prepayment-driven demand: Table 2, Figure 3, Table 3, Table 4	10
5.3.1	Baseline time-series model (Table 2)	10
5.3.2	Figure 3 and Table 3: mortgage rates vs. federal funds target	11
5.3.3	Table 4: panel identification using coupon cohorts (a “within-month” design)	12
5.4	Government guarantee and regulation: Figure 4 and Table 5	13
5.4.1	Key idea	13
5.4.2	Figure 4 (event windows)	13
5.4.3	Table 5 regression model	13
5.5	Further evidence: Table 6 (dealer dispersion), Table 7 (seasoned MBS), Tables 8–9 (holdings)	15
5.5.1	Table 6: Dealer prepayment forecast dispersion	15
5.5.2	Table 7: Seasoned MBS	15
5.5.3	Tables 8–9: Investors’ holdings and the prepayment channel	16

6	Integrated takeaway: one coherent mechanism across prices, issuance, and institutions	19
7	Conclusion (What the paper establishes)	19
7.1	Core message	19
7.2	Main empirical conclusions to remember	19
8	Intuition (Why these results are economically natural)	20
8.1	Safe-asset pricing: a liquidity-services view	20
8.2	Why mortgage rates matter: the prepayment-driven demand channel	21
8.3	Why policy events move convenience premia	21
8.4	A final “big picture” intuition	21
9	Short concluding paragraph	22

1 What the paper is trying to do

This paper asks a precise asset-pricing question: *Do agency mortgage-backed securities (agency MBS) behave like long-term safe assets, and through which economic channels does their “safe-asset” status manifest in prices and quantities?*

The organizing idea is the **convenience premium** (or liquidity/safety premium). Safe assets deliver non-pecuniary services (liquidity, pledgeability, regulatory benefits, and informational simplicity). These services raise investors’ willingness to hold the asset, pushing *yields down* relative to otherwise comparable assets. The paper builds a simple demand–supply framework that links:

- the **convenience premium** of agency MBS (measured by yield spreads), and
- the **issuance** of newly issued agency MBS,

to demand shifters (especially those tied to **prepayment risk**), and to institutional shifts that affect **government backing** and **regulatory treatment**.

The empirical work then maps the model objects into observables:

- convenience premium $\leftrightarrow$ yield spreads against Treasury and AAA corporate benchmarks;
- supply response $\leftrightarrow$ issuance of newly issued “production-coupon” FN30y MBS;
- MBS-specific demand shifter $\lambda_t^M \leftrightarrow$ mortgage-rate-driven prepayment incentive (plus complementary evidence from dealer prepayment-forecast dispersion);
- policy-based shifts in safety/regulation $\leftrightarrow$ the 2008 conservatorship event and the 2013 LCR regulatory haircut differential.

2 Theoretical framework: demand for safe assets, pricing, and issuance

2.1 Preferences and the safe-asset “liquidity” aggregator

A representative agent receives an endowment stream $\{A_t\}$ and maximizes

$$\mathbb{E}_0 \left[\sum_{t=1}^{\infty} \beta^t u(C_t) \right], \quad \beta \in (0, 1), \quad (1)$$

where the effective consumption index is

$$C_t \equiv c_t + \gamma_t v(Q_t), \quad (1)$$

and the agent’s *real liquidity holdings* are aggregated as

$$Q_t \equiv Q_t^B + \lambda_t^M Q_t^M. \quad (2)$$

Interpretation and assumptions:

- c_t is physical consumption.
- $v(\cdot)$ captures the utility value of holding safe/liquid assets; assume $v' > 0$, $v'' < 0$.
- $\gamma_t > 0$ scales the aggregate demand for all safe assets (flight-to-safety, overall liquidity demand, etc.).
- Q_t^B is real holdings of Treasury securities; Q_t^M is real holdings of agency MBS.
- $\lambda_t^M \in (0, 1)$ is the **relative liquidity/safety service intensity** of agency MBS relative to Treasuries.
 - Higher λ_t^M means agency MBS behave more like Treasuries in providing safe-asset services.

- Economically, λ_t^M is lowered by MBS-specific frictions (especially prepayment-related complexity/uncertainty), and raised by stronger government backing and favorable regulatory treatment.
- A regularity condition used for comparative statics is:

$$v'(Q) + Q v''(Q) \geq 0, \quad (2)$$

which ensures certain monotonicity properties (it prevents marginal convenience from falling “too fast”).

2.2 Budget constraint and Euler equations

Let Z_t be holdings of a bond that *does not* produce liquidity services (a “pure” bond), and let P_t^Z, P_t^B, P_t^M be (real) prices. The period budget constraint is

$$c_t + P_t^Z Z_t + P_t^B Q_t^B + P_t^M Q_t^M = A_t + Z_{t-1} + Q_{t-1}^B + Q_{t-1}^M. \quad (3)$$

The Euler equation for the non-service bond Z is standard:

$$P_t^Z = \mathbb{E}_t \left[\beta \frac{u'(C_{t+1})}{u'(C_t)} P_{t+1}^Z \right]. \quad (4)$$

For Treasuries and MBS, the key modification is that holdings provide convenience services. The Euler equations become:

$$P_t^M = \lambda_t^M \gamma_t v'(Q_t) P_t^M + \beta \mathbb{E}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} P_{t+1}^M \right], \quad (5)$$

$$P_t^B = \gamma_t v'(Q_t) P_t^B + \beta \mathbb{E}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} P_{t+1}^B \right]. \quad (6)$$

Economic meaning: a marginal unit of MBS delivers only a fraction λ_t^M of the safe-asset service provided by a Treasury dollar.

2.3 Convenience premiums: one-period yields and a log approximation

To make the link to yields transparent, consider one-period bonds so that $P_{t+1}^Z = P_{t+1}^B = P_{t+1}^M = 1$. Define the one-period yield r_t on the non-service bond by

$$e^{-r_t} \equiv P_t^Z = \mathbb{E}_t \left[\beta \frac{u'(C_{t+1})}{u'(C_t)} \right]. \quad (3)$$

Then the MBS price satisfies

$$e^{-r_t^M} \equiv P_t^M = \frac{\mathbb{E}_t [\beta u'(C_{t+1})/u'(C_t)]}{1 - \lambda_t^M \gamma_t v'(Q_t)} \approx e^{\lambda_t^M \gamma_t v'(Q_t)} \mathbb{E}_t \left[\beta \frac{u'(C_{t+1})}{u'(C_t)} \right]. \quad (7)$$

Taking logs under the approximation delivers the central mapping:

$$r_t - r_t^M \approx \lambda_t^M \gamma_t v'(Q_t), \quad (8)$$

and similarly for Treasuries:

$$r_t - r_t^B \approx \gamma_t v'(Q_t). \quad (9)$$

Interpretation.

- $r_t - r_t^M$ is the **convenience premium** of agency MBS relative to a non-service benchmark.
- $r_t - r_t^B$ is the **convenience premium** of Treasuries.
- The Treasury–MBS yield spread is

$$r_t^B - r_t^M = (r_t - r_t^M) - (r_t - r_t^B) \approx (\lambda_t^M - 1)\gamma_t v'(Q_t), \quad (13)$$

which is typically negative because $\lambda_t^M < 1$, meaning MBS yields are above Treasuries on average (an “inconvenience discount” relative to Treasuries).

2.4 Supply side: issuance by MBS intermediaries

Agency MBS are supplied (issued) by intermediaries. The paper represents issuance decisions in reduced form as a monopoly supplier choosing Q_t^M to maximize

$$\Pi_t = (r_t - r_t^M)Q_t^M - \kappa(Q_t^M), \quad (4)$$

with $\kappa' > 0$ and $\kappa'' > 0$ (convex marginal issuance cost). The first-order condition is

$$r_t - r_t^M = \kappa'(Q_t^M). \quad (5)$$

Let $\phi(\cdot) \equiv (\kappa')^{-1}(\cdot)$ denote the supply function. Then

$$Q_t^M = \phi(r_t - r_t^M), \quad \phi' > 0. \quad (10)$$

2.5 Equilibrium system

Combine demand-side pricing with supply:

$$r_t - r_t^M = \lambda_t^M \gamma_t v'(Q_t), \quad (11)$$

$$r_t - r_t^B = \gamma_t v'(Q_t), \quad (12)$$

$$Q_t = Q_t^B + \lambda_t^M Q_t^M, \quad (2 \text{ revisited})$$

$$Q_t^M = \phi(r_t - r_t^M). \quad (10 \text{ revisited})$$

Substituting, define $x_t \equiv r_t - r_t^M$. Then

$$x_t = \lambda_t^M \gamma_t v'(Q_t^B + \lambda_t^M \phi(x_t)). \quad (6)$$

This single implicit equation is enough to generate comparative statics and the empirical predictions.

3 Comparative statics: propositions and proofs (fully spelled out)

3.1 Proposition 1: changes in MBS-specific safety/liquidity λ_t^M

Proposition 1 (Effects of λ_t^M). *Assume $v' > 0$, $v'' < 0$, $\phi' > 0$, and $v'(Q) + Qv''(Q) \geq 0$. Holding Q_t^B and γ_t fixed, an increase in λ_t^M implies:*

1. *MBS convenience premium increases: $d(r_t - r_t^M)/d\lambda_t^M > 0$.*
2. *MBS issuance increases: $dQ_t^M/d\lambda_t^M > 0$.*

3. The MBS yield relative to Treasury decreases: $d(r_t^M - r_t^B)/d\lambda_t^M < 0$ (equivalently, $d(r_t^B - r_t^M)/d\lambda_t^M > 0$).

Proof. Let $x \equiv r_t - r_t^M$ and $\lambda \equiv \lambda_t^M$, $\gamma \equiv \gamma_t$, and define

$$Q(x, \lambda) \equiv Q^B + \lambda\phi(x).$$

Define the equilibrium condition as

$$F(x, \lambda) \equiv x - \lambda\gamma v'(Q(x, \lambda)) = 0.$$

Compute the partial derivatives (holding the other argument constant):

$$\frac{\partial F}{\partial x} = 1 - \lambda\gamma v''(Q) \cdot \frac{\partial Q}{\partial x} = 1 - \lambda\gamma v''(Q) \cdot (\lambda\phi'(x)) = 1 + \lambda^2\gamma(-v''(Q))\phi'(x) > 0.$$

Hence the implicit function theorem applies.

Next,

$$\frac{\partial F}{\partial \lambda} = -\gamma v'(Q) - \lambda\gamma v''(Q) \cdot \frac{\partial Q}{\partial \lambda} = -\gamma v'(Q) - \lambda\gamma v''(Q) \cdot \phi(x).$$

Therefore,

$$\frac{dx}{d\lambda} = -\frac{F_\lambda}{F_x} = \frac{\gamma(v'(Q) + \lambda v''(Q)\phi(x))}{F_x}.$$

Because $Q = Q^B + \lambda\phi(x)$, we can rewrite

$$v'(Q) + \lambda v''(Q)\phi(x) = (v'(Q) + Qv''(Q)) + (-v''(Q))Q^B.$$

By assumption $v'(Q) + Qv''(Q) \geq 0$ and $-v''(Q) > 0$, so the expression is strictly positive. Since $F_x > 0$, we obtain $dx/d\lambda > 0$, proving Part (1).

Part (2): $Q^M = \phi(x)$ and $\phi' > 0$, so

$$\frac{dQ^M}{d\lambda} = \phi'(x)\frac{dx}{d\lambda} > 0.$$

Part (3): using (6) and (6)–(13), the Treasury–MBS yield spread is

$$r_t^B - r_t^M = (\lambda - 1)\gamma v'(Q).$$

Differentiate:

$$\frac{d(r_t^B - r_t^M)}{d\lambda} = \gamma v'(Q) + (\lambda - 1)\gamma v''(Q)\frac{dQ}{d\lambda}.$$

Because $\lambda - 1 < 0$, $v''(Q) < 0$, and $dQ/d\lambda > 0$ (since $Q = Q^B + \lambda\phi(x)$ and both $\phi(x)$ and x increase with λ), the second term is positive. Thus the whole derivative is strictly positive. Equivalently, $d(r_t^M - r_t^B)/d\lambda < 0$. $\square$

3.2 Proposition 2: changes in overall safe-asset demand γ_t

Proposition 2 (Effects of γ_t). Assume $v' > 0$, $v'' < 0$, and $\phi' > 0$. Holding Q_t^B and λ_t^M fixed, an increase in γ_t implies:

1. MBS convenience premium increases: $d(r_t - r_t^M)/d\gamma_t > 0$.
2. MBS issuance increases: $dQ_t^M/d\gamma_t > 0$.

3. *Treasury–MBS spread decreases*: $d(r_t^B - r_t^M)/d\gamma_t < 0$ (becomes more negative because $\lambda_t^M < 1$).

Proof. Again set $x \equiv r_t - r_t^M$ and define

$$F(x, \gamma) \equiv x - \lambda\gamma v'(Q(x, \lambda)) = 0, \quad Q(x, \lambda) = Q^B + \lambda\phi(x).$$

We already computed $F_x = 1 + \lambda^2\gamma(-v''(Q))\phi'(x) > 0$.

Now

$$F_\gamma = -\lambda v'(Q).$$

Therefore,

$$\frac{dx}{d\gamma} = -\frac{F_\gamma}{F_x} = \frac{\lambda v'(Q)}{F_x} > 0,$$

proving Part (1). Then $Q^M = \phi(x)$ with $\phi' > 0$ implies Part (2).

For Part (3), write

$$r_t^B - r_t^M = (\lambda - 1)\gamma v'(Q).$$

Differentiate w.r.t. γ :

$$\frac{d(r_t^B - r_t^M)}{d\gamma} = (\lambda - 1)v'(Q) + (\lambda - 1)\gamma v''(Q) \frac{dQ}{d\gamma}.$$

Since $dQ/d\gamma = \lambda\phi'(x)dx/d\gamma$, substitute $dx/d\gamma = \lambda v'(Q)/F_x$:

$$\frac{dQ}{d\gamma} = \lambda\phi'(x) \cdot \frac{\lambda v'(Q)}{F_x} = \frac{\lambda^2\phi'(x)v'(Q)}{F_x}.$$

Hence

$$\frac{d(r_t^B - r_t^M)}{d\gamma} = (\lambda - 1)v'(Q) \left(1 + \gamma v''(Q) \frac{\lambda^2\phi'(x)}{F_x} \right).$$

Let $A \equiv \lambda^2\gamma(-v''(Q))\phi'(x) > 0$, so $F_x = 1 + A$ and

$$1 + \gamma v''(Q) \frac{\lambda^2\phi'(x)}{F_x} = 1 - \frac{A}{1 + A} = \frac{1}{1 + A} > 0.$$

Therefore the bracket is positive. Since $(\lambda - 1) < 0$ and $v'(Q) > 0$, the derivative is strictly negative: $d(r_t^B - r_t^M)/d\gamma < 0$. $\square$

3.3 A crowding-out implication (Treasury supply)

Although not formally stated as a proposition above, the equilibrium equation (6) makes clear that a larger exogenous Treasury supply Q_t^B raises total liquidity holdings Q_t , reducing marginal convenience $v'(Q_t)$ (because $v'' < 0$). This lowers $x_t = r_t - r_t^M$ and thus lowers issuance $Q_t^M = \phi(x_t)$: a **crowding-out effect** of Treasuries (and similarly, other outstanding safe assets) on newly issued agency MBS.

4 From theory to empirics: how the model objects are measured

4.1 Key observables and their model interpretation

The paper focuses on **newly issued Fannie Mae 30-year production-coupon MBS** (denoted FN30y). The main price-based objects are yield spreads:

$$\text{Treasury-FN30y}_t \equiv y_t^{\text{Treas}} - y_t^{\text{FN30y}}, \quad (7)$$

$$\text{AAA-FN30y}_t \equiv y_t^{\text{AAA}} - y_t^{\text{FN30y}}, \quad (8)$$

$$\text{AAA-Treasury}_t \equiv y_t^{\text{AAA}} - y_t^{\text{Treas}}. \quad (9)$$

Mapping to the model:

- AAA-FN30y is a proxy for $r_t - r_t^M$ (MBS convenience premium against a non-convenience benchmark).
- AAA-Treasury is a proxy for $r_t - r_t^B$ (long-term Treasury convenience premium).
- Treasury-FN30y is a proxy for $r_t^B - r_t^M$ (Treasury-MBS spread; often negative because Treasuries are “more convenient”).

The primary quantity-based object is:

$$\text{Issuance}_t \equiv \text{monthly issuance amount of production-coupon FN30y MBS}, \quad (10)$$

which matches the model prediction that issuance responds positively to the convenience premium via $Q_t^M = \phi(r_t - r_t^M)$.

4.2 Prepayment-driven demand: the economic mechanism and empirical proxy

The paper’s central MBS-specific demand shifter is λ_t^M , and the prepayment channel argues that higher refinancing incentives increase prepayment complexity/uncertainty, making MBS less “Treasury-like.” The paper summarizes this channel as

$$\lambda_t^M = -(\text{prepayment model complexity}_t) \times f(m_t), \quad f(\cdot) > 0 \text{ increasing}, \quad (14)$$

where m_t is the mortgage rate of the newly issued MBS.

A microfoundation (following a reduced-form prepayment specification) writes the future prepayment rate of a time- t issued mortgage as

$$p_s^t = x_s + y_s \cdot \max\{0, m_t - m_s\}, \quad s \geq t, \quad (15)$$

where $m_t - m_s$ is the **moneyness** (refinancing incentive) and y_s is a “rate response” sensitivity. For two issuance dates t_1, t_2 , this yields

$$\begin{aligned} p_s^{t_1} &= x_s + y_s \cdot \max\{0, m_{t_1} - m_s\}, \\ p_s^{t_2} &= x_s + y_s \cdot \max\{0, m_{t_2} - m_s\}, \end{aligned} \quad (16)$$

so differences in the *at-issuance* mortgage rate (m_{t_1} vs. m_{t_2}) directly shift the entire prepayment profile. This motivates the empirical focus on the observed mortgage rate (PMMS) as the main proxy for prepayment-driven demand variation in normal times.

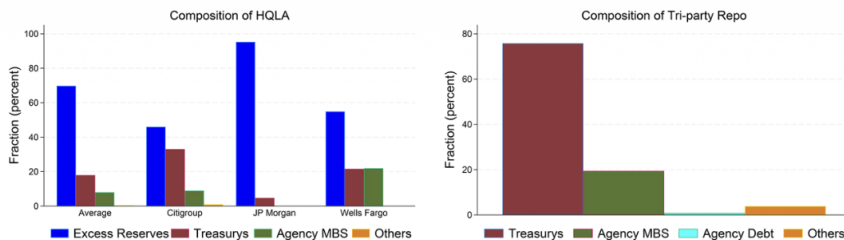


Figure 1: HQLA and repo

5 Figures and tables: what each one shows and how I interpret it

5.1 Figure 1: HQLA and repo (why agency MBS matter as “safe assets”)

What the figure contains.

- Left panel: fractions of excess reserves, Treasuries, agency MBS, and other securities in HQLA (reported for several large banks and an average).
- Right panel: composition of tri-party repo collateral (Treasuries, agency MBS, agency debt, and other investment-grade assets).

My interpretation. This figure motivates the paper’s premise: agency MBS are not a niche asset class. They appear prominently in:

- banks’ high-quality liquid asset buffers (HQLA), and
- the collateral universe used in secured funding markets.

Therefore, even if agency MBS are not as “perfectly” safe/liquid as Treasuries, their scale and institutional roles imply that studying their convenience premium is economically important.

5.2 Figure 2 and Table 1: measuring and quantifying the MBS convenience premium

Figure 2 (time series) shows:

- left panel: monthly series of AAA–FN30y, Treasury–FN30y, and AAA–Treasury;
- right panel: monthly issuance amount of production-coupon FN30y.

Table 1 (summary statistics) provides:

- average levels and dispersion of the yield spreads and issuance.

My interpretation.

- The mean AAA–FN30y is large and positive, indicating that agency MBS yields are *below* AAA corporate yields on average: a sizable convenience premium relative to AAA corporates.
- The mean Treasury–FN30y is negative, indicating that Treasuries still enjoy a stronger convenience premium (lower yields) than agency MBS: agency MBS are “safe” but not as “convenient” as Treasuries.
- The paper emphasizes a magnitude comparison: the agency MBS convenience premium is *more than half* of the long-term Treasury convenience premium proxy (AAA–Treasury).
- Issuance is economically large and time-varying, which is critical because the model predicts that issuance responds to the convenience premium.

Table 1: Summary statistics of yield spreads and MBS issuance amount (monthly, Oct 2003–Dec 2022)

	Mean	SD	Min	P25	P50	P75	Max
Treasury–FN30y	-27.61	24.13	-113.02	-36.26	-24.94	-14.10	21.94
AAA–FN30y	38.50	46.44	-50.19	18.41	32.42	43.54	313.32
AAA–Treasury	66.11	52.12	25.75	46.13	52.59	62.88	374.63
FN30y issuance	20.95	12.66	5.91	13.04	16.96	23.89	76.97

Notes: Yield spreads are in basis points. Issuance is in billions of dollars. “Treasury–FN30y” is Treasury yield minus newly issued FN30y yield; negative values indicate FN30y yields above Treasuries on average.

5.3 Prepayment-driven demand: Table 2, Figure 3, Table 3, Table 4

5.3.1 Baseline time-series model (Table 2)

Empirical specification. The baseline time-series regressions are of the form

$$Y_t = \alpha + \beta \text{PMMS}_t + \delta_1 \text{VIX}_t + \delta_2 \log(\text{Debt}/\text{GDP})_t + \delta_3 \log(\text{Seasoned MBS}/\text{GDP})_t + \varepsilon_t, \quad (11)$$

with Y_t being:

- Treasury–FN30y_t (Treasury–MBS spread),
- AAA–FN30y_t (MBS convenience premium vs. AAA corporates),
- Issuance_t (monthly FN30y issuance).

Model-based sign prediction. Higher mortgage rates increase refinancing incentives and prepayment concerns, lowering λ_t^M and thus lowering $r_t - r_t^M$ and issuance. Therefore, $\beta < 0$ is expected in all three regressions.

Table 2: Mortgage rate and agency MBS convenience premium

	(1) Treasury–FN30y	(2) AAA–FN30y	(3) FN30y issuance
PMMS	-14.473*** (-4.264)	-23.735*** (-4.081)	-9.118*** (-5.589)
VIX	-1.257*** (-4.905)	3.207*** (2.673)	0.217 (1.481)
log(Debt/GDP)	-36.665** (-2.518)	-83.054*** (-3.574)	10.820** (1.975)
log(Seasoned MBS/GDP)	-7.341 (-0.350)	-39.924 (-1.597)	-29.813*** (-3.492)
Intercept	205.311** (2.492)	379.350*** (3.092)	-21.594 (-0.745)
N	231	231	231
R^2	0.365	0.481	0.485

Notes: Monthly time-series regressions. t-statistics are in parentheses (Newey–West). Stars denote significance at 10/5/1%.

My interpretation of Table 2.

- The negative PMMS coefficients match the prepayment channel: higher mortgage rates coincide with lower convenience premium (vs. AAA), a more negative Treasury–FN30y spread, and lower issuance.

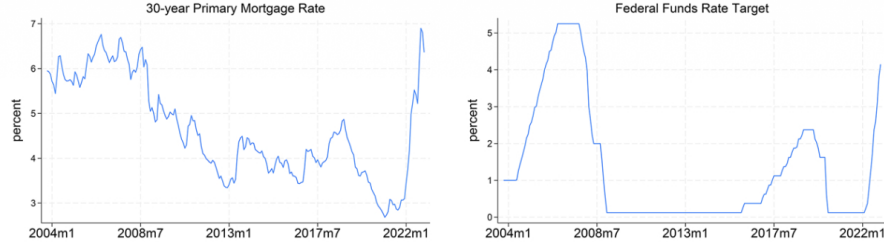


Figure 2: Figure 3 Monthly series of mortgage rate and federal funds target rate

- VIX has opposite signs for Treasury–FN30y (negative) and AAA–FN30y (positive), consistent with a “demand for safe assets” force (γ_t) that boosts convenience values broadly but strengthens Treasuries more than MBS.
- The supply controls matter: in particular, more outstanding seasoned MBS is strongly associated with lower new issuance, consistent with crowding out from alternative safe assets.

5.3.2 Figure 3 and Table 3: mortgage rates vs. federal funds target

Figure 3 (time series) shows:

- mortgage rates and the federal funds target co-move over the long sample, especially around tightening/easing cycles.

Table 3 (regressions) is used as a diagnostic:

- Column (1): $PMMS_t$ regressed on FFR_t (strong positive association).
- Columns (2)–(4): replace $PMMS$ with FFR in the baseline regressions to see whether results could be mechanically driven by general interest-rate conditions rather than prepayment-specific demand.

Table 3: Federal funds rate and agency MBS convenience premium

	(1) PMMS	(2) Treasury–FN30y	(3) AAA–FN30y	(4) FN30y issuance
FFR	0.507*** (12.293)	–8.621*** (–4.950)	–15.192*** (–4.260)	–1.837* (–1.672)
VIX		–1.767*** (–8.261)	2.328** (2.201)	0.043 (0.290)
log(Debt/GDP)		–5.662 (–0.383)	–32.632* (–1.691)	31.791*** (3.664)
log(Seasoned MBS/GDP)		–32.569 (–1.577)	–85.232*** (–2.909)	–32.287*** (–3.236)
Intercept	3.973*** (29.713)	1.327 (0.016)	43.983 (0.425)	–147.243*** (–3.235)
N	231	231	231	231
R^2	0.540	0.398	0.522	0.345

Notes: Column (1) links mortgage rates (PMMS) to the federal funds target (FFR). Columns (2)–(4) use FFR as the main regressor in place of PMMS. t-statistics are in parentheses (Newey–West). Stars denote significance at 10/5/1%.

My interpretation of Table 3.

- Mortgage rates respond strongly to FFR (Column 1).

A. Sample summary				
Issuance rank	Moneyness	FN30y issuance	AAA-FN30y	Treasury-FN30y
1	0.329	20.406	43.07	-23.48
2	0.468	10.692	41.53	-25.01
3	0.593	3.814	37.57	-29.80
B. Regressions				
	ALL		Issuance rank ≤ 2	
	Treasury-FN30y	FN30y issuance	Treasury-FN30y	FN30y issuance
Mortgage rate	-8.541*** (-6.243)	-6.123*** (-6.541)	-6.005** (-2.466)	-8.270*** (-3.454)
N	644	644	434	434
R^2	.906	.481	.936	.646
Time FE	Yes	Yes	Yes	Yes

Figure 3: Table 4 Panel analysis of the MBS convenience premium and issuance

- If the safe-asset convenience premium were primarily driven by a general “opportunity cost of money” channel, I would expect higher short-term rates to be associated with *higher* convenience premia for safe assets (in particular, a stronger positive relation with AAA-FN30y). Instead, AAA-FN30y falls with FFR, aligning more naturally with the prepayment logic operating through mortgage-rate conditions.
- The joint pattern supports the paper’s claim that the main driver is MBS-specific prepayment-related demand rather than generic interest-rate-level effects.

5.3.3 Table 4: panel identification using coupon cohorts (a “within-month” design)

Panel construction. Each month t contains several newly issued coupon cohorts. The paper ranks cohorts by issuance and focuses on the top 3 (or top 2) cohorts. It defines **moneyness** for newly issued MBS as:

$$\text{moneyness}_{it} = (\text{coupon rate})_{it} - (\text{current-coupon rate})_t.$$

Panel regression model. For cohort i in month t :

$$Y_{it} = D_t + \beta \text{MortgageRate}_{it} + \varepsilon_{it}, \quad (12)$$

where D_t are time fixed effects and Y_{it} is either $\text{Treasury-FN30y}_{it}$ or issuance_{it} .

Why this matters. Time fixed effects absorb aggregate shocks (including any time-varying γ_t). Identification comes from cross-sectional differences across cohorts *within the same month*, aligning closely with the model logic that mortgage-rate-driven prepayment incentives vary across cohorts at a given time.

My interpretation of Table 4.

- The negative mortgage-rate coefficients persist even with time fixed effects, which is strong evidence that the prepayment-driven channel is not merely a time-series artifact.
- The within-month cross-sectional design is a clean bridge between the theory (where λ_t^M can vary by MBS “type”) and the data (cohorts differ in coupon/moneyness and thus prepayment exposure).

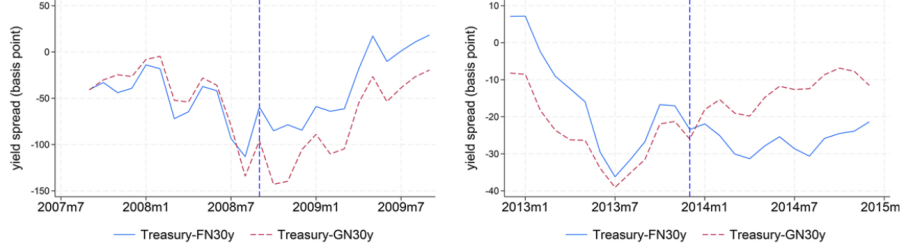


Figure 4: Policy events and convenience premium

5.4 Government guarantee and regulation: Figure 4 and Table 5

5.4.1 Key idea

Policy events shift MBS safety/regulatory status, which shifts λ_t^M independently of mortgage-rate-driven prepayment conditions (or at least provides quasi-exogenous variation).

5.4.2 Figure 4 (event windows)

Figure 4 plots Treasury–FN30y and Treasury–GN30y around:

- the 2008 conservatorship (event time Sep 2008),
- the 2013 LCR period (event time Dec 2013).

The visual suggests relative changes in FN vs GN convenience around these events.

5.4.3 Table 5 regression model

In each event window, the empirical model is:

$$\text{Spread}_t = \alpha + \beta \text{PostEvent}_t + \theta \text{PMMS}_t + \text{controls}_t + \varepsilon_t, \quad (13)$$

run separately for Treasury–FN30y, Treasury–GN30y, and their difference GN30y–FN30y.

Economic interpretation of the difference regression. GN30y is used as a control group:

- GN30y has an explicit government guarantee already, so it should not respond to FN-specific conservatorship.
- LCR haircuts differ: FN (and Freddie) face a haircut while GN and Treasuries face zero haircut, shifting relative regulatory convenience.

Table 4: Paper's Table 5 Effects of policy events

	A. Post-conservatorship			B. Post-LCR			C. Full sample
	Treasury–FN30y	Treasury–GN30y	GN30y–FN30y	Treasury–FN30y	Treasury–GN30y	GN30y–FN30y	Treasury–GN30y
Post-event	43.19*** (2.77)	10.12 (0.46)	33.06*** (2.85)	−3.43 (−1.48)	4.00 (1.17)	−7.43*** (−3.04)	—
PMMS	−24.51*** (−2.58)	−43.03*** (−4.14)	18.52*** (3.41)	−24.47*** (−8.18)	−19.62*** (−6.06)	−4.85** (−2.48)	−16.070*** (−3.027)
VIX	−0.80** (−2.19)	−1.14*** (−3.71)	0.33** (2.11)	−0.48 (−0.78)	0.04 (0.06)	−0.52 (−1.28)	−1.880*** (−7.124)
log(Debt/GDP)	−110.15* (−1.85)	−200.31** (−2.25)	90.16** (1.97)	534.59*** (6.97)	366.93*** (5.33)	167.66*** (4.69)	−59.668*** (−3.895)
log(Seasoned MBS/GDP)	−553.63*** (−6.05)	−538.67*** (−4.73)	−14.96 (−0.28)	227.02*** (3.76)	−191.17* (−1.68)	418.19*** (4.60)	14.108 (0.951)
Intercept	−364.24* (−1.68)	113.18 (0.43)	−477.42*** (−3.31)	−1,966.51*** (−5.96)	−1,708.77*** (−5.57)	−257.74 (−1.46)	342.931*** (3.836)
N	25	25	25	25	25	25	219
R^2	0.81	0.86	0.91	0.86	0.82	0.97	0.468

Notes: Panel A uses a short window around the 2008 conservatorship; Panel B uses a short window around the 2013 LCR-related period; Panel C reports the full-sample baseline for Treasury–GN30y. t-statistics are in parentheses (Newey–West). Stars denote significance at 10/5/1%.

<i>A. Sample summary</i>					
Issuance rank	Moneyiness	FN30y issuance	Treasury-FN30y	AAA-FN30y	Dealer dispersion
1	0.366	18.389	-24.89	44.56	8.38
2	0.540	9.632	-25.84	41.58	8.77
3	0.630	3.690	-31.19	39.12	8.45

<i>B. Regressions</i>				
	Dealer dispersion	FN30y issuance	Treasury-FN30y	Treasury-FN30y
Mortgage rate	2.832*** (4.228)			-9.433*** (-5.739)
Dealer dispersion		-0.069 (-0.716)	-0.449*** (-2.919)	-0.108 (-0.712)
N	573	573	573	573
R ²	.657	.281	.902	.922
Time FE	Yes	Yes	Yes	Yes

Figure 5: Paper’s Table 6 Dealer prepayment forecast dispersion

My interpretation of Table 5.

- **Conservatorship:** the post-event dummy is strongly positive for Treasury–FN30y but weak for Treasury–GN30y; the difference GN30y–FN30y rises significantly. This supports a discrete upward shift in FN safety/convenience due to explicit government backing.
- **LCR:** the key result is a significantly negative coefficient on post-LCR in GN30y–FN30y, meaning FN convenience declines relative to GN after the LCR haircut differential (FN haircuts vs. zero for GN/Treasuries).
- This is precisely the paper’s identification logic: it isolates institutional channels affecting λ_t^M from general market-wide drivers.

5.5 Further evidence: Table 6 (dealer dispersion), Table 7 (seasoned MBS), Tables 8–9 (holdings)

5.5.1 Table 6: Dealer prepayment forecast dispersion

Idea. If prepayment complexity/uncertainty is important, I should observe:

- higher mortgage rates $\Rightarrow$ higher dispersion in dealer prepayment forecasts;
- higher dispersion $\Rightarrow$ lower convenience premium and/or lower issuance.

My interpretation of Table 6.

- Mortgage rates predict dealer dispersion positively, supporting the channel that higher refinancing incentives raise prepayment-model complexity/uncertainty.
- Higher dispersion is associated with a lower Treasury–FN30y spread (more negative), consistent with reduced relative convenience of MBS.
- When mortgage rate and dispersion enter jointly, dispersion loses significance, suggesting mortgage rate is the main sufficient statistic for this dimension of prepayment-driven demand in the baseline analysis.

5.5.2 Table 7: Seasoned MBS

Goal. Verify that the prepayment mechanism is not confined to newly issued production-coupon MBS.

Construction. Exclude the newly issued cohorts and form a panel of seasoned coupon cohorts. Define moneyiness for seasoned MBS as the refinancing incentive implied by the relationship between

the underlying mortgage rate and current PMMS (conceptually, when current rates fall, seasoned high-coupon MBS become more in-the-money).

Table 5: Paper’s Table 7 Seasoned MBS

A. Sample summary					
Coupon cohort	Begin	End	Moneyiness	AAA–FN30y	Treasury–FN30y
2	Jan. 2013	Apr. 2022	-1.030	-6.08	-61.41
2.5	Sep. 2011	Apr. 2022	-0.538	27.73	-26.91
3	Jan. 2009	Apr. 2021	-0.432	42.88	-22.00
3.5	Jan. 2009	Feb. 2022	0.393	53.39	-18.09
4	Oct. 2003	Mar. 2022	-0.303	27.01	-46.11
4.5	Oct. 2003	Apr. 2022	0.698	28.23	-36.83
5	Aug. 2004	Apr. 2022	1.799	20.16	-37.27
5.5	Jul. 2006	Apr. 2022	2.487	-0.20	-59.16
6	Mar. 2005	Apr. 2022	2.825	-22.19	-84.80
6.5	Oct. 2003	Apr. 2022	2.710	-23.68	-91.68
7	Oct. 2003	Apr. 2022	2.757	-13.42	-84.60
7.5	Oct. 2003	Apr. 2022	2.763	-25.28	-102.46
8	Oct. 2003	Apr. 2022	2.869	-67.90	-152.49
B. Regression results					
	AAA–FN30y		Treasury–FN30y		
Moneyiness	−10.059*** (−5.010)		−6.271*** (−2.820)		
VIX	2.236*** (5.524)		−2.299*** (−5.165)		
log(Debt/GDP)	1.841* (1.799)		57.201*** (8.035)		
N	1,550		1,550		
R^2	0.399		0.343		
Coupon FE	Yes		Yes		

Notes: Panel regressions for seasoned coupon cohorts with coupon fixed effects. t-statistics are in parentheses. Stars denote significance at 10/5/1%.

My interpretation of Table 7.

- Moneyiness enters negatively for both spreads: when refinancing incentives are stronger, the convenience premium weakens, consistent with higher prepayment-related frictions.
- This confirms that the prepayment channel applies beyond newly issued MBS.
- Conceptually important nuance: a given change in current mortgage rates can shift the moneyiness of seasoned and newly issued cohorts differently, implying distributional effects within the MBS universe.

5.5.3 Tables 8–9: Investors’ holdings and the prepayment channel

Idea. If MBS convenience varies with mortgage-rate-driven prepayment conditions, then investors’ *portfolio shares* in agency MBS relative to Treasuries should also respond.

My interpretation of Table 8.

- There is strong heterogeneity in the *relative concentration* in agency MBS across investor types (e.g., GSEs and depositories are much more MBS-heavy than foreign investors).

Table 6: Paper’s Table 8 Summary of holdings (quarterly, 1995:Q4–2022:Q4)

	Treasury		MBS		MBS/Treasury	
	Mean	SD	Mean	SD	Mean	SD
GSE	72	65	526	309	22.86	25.09
U.S. depository	329	358	1,533	747	7.79	4.79
Credit union	21	16	133	72	7.32	2.77
MMF	165	161	398	223	4.38	3.20
Life insurance	138	61	348	64	2.95	1.15
Mutual funds	480	434	528	207	1.87	1.04
P&C insurance	84	22	111	24	1.41	0.39
Private pension	225	129	197	26	1.17	0.60
Local gov	450	315	405	137	1.11	0.38
Local gov retirement	210	73	157	60	0.85	0.48
Rest of the world	3,280	2,152	869	412	0.33	0.19
Total	8,910	6,265	7,037	2,433	1.04	0.46

Notes: The table summarizes quarterly holdings of Treasury securities and agency MBS and the ratio of MBS to Treasury holdings for major investor groups.

- This heterogeneity sets up a meaningful cross-group prediction: groups that are structurally MBS-heavy should respond more strongly in their MBS/Treasury ratio to changes in mortgage-rate-driven demand conditions.

Table 7: Paper's Table 9 Mortgage rate and investors' holdings: regressions of MBS/Treasury ratio

	GSE	Depository	Credit union	MMF	Life	MF	P&C	Private pension	Local gov	Local gov retirement	RoW
PMMS	-7.14* (-1.90)	-2.05*** (-2.96)	-1.39*** (-5.66)	-0.93** (-2.21)	-0.20** (-2.41)	-0.23*** (-2.97)	-0.16*** (-3.74)	0.0003 (0.008)	-0.06 (-1.55)	-0.19** (-2.27)	-0.06** (-2.03)
log(Debt/GDP)	-83.63*** (-7.47)	-17.28*** (-8.05)	-10.59*** (-15.09)	-11.09*** (-8.22)	-3.92*** (-14.35)	-3.72*** (-18.72)	-1.46*** (-10.57)	-1.06*** (-5.85)	-1.39*** (-13.38)	-1.80*** (-6.83)	-0.68*** (-7.58)
log(MBS/GDP)	3.52 (0.33)	2.89 (1.57)	4.46*** (5.12)	2.29** (2.05)	0.55** (2.27)	0.69*** (3.02)	0.73*** (4.69)	-0.51*** (-4.72)	0.39*** (3.60)	0.41** (2.17)	0.12 (1.46)
Intercept	398.07*** (7.17)	91.51*** (8.53)	63.14*** (18.15)	56.61*** (8.19)	20.36*** (15.28)	18.80*** (18.82)	9.08*** (12.87)	4.65*** (5.22)	7.51*** (14.80)	9.58*** (7.37)	3.55*** (7.97)
N	109	109	109	109	109	109	109	109	109	109	109
R^2	0.81	0.67	0.79	0.71	0.93	0.89	0.80	0.90	0.91	0.73	0.74

Notes: Each column is a separate regression of the investor group's MBS/Treasury holdings ratio on PMMS and supply controls. t-statistics are in parentheses. Stars denote significance at 10/5/1%.

My interpretation of Table 9.

- For most investor groups, PMMS enters negatively: higher mortgage rates predict lower MBS/Treasury holdings ratios, consistent with reduced demand for MBS when prepayment concerns rise.
- The pattern of magnitudes is consistent with heterogeneous exposure: investor groups that are more MBS-heavy (Table 8) tend to exhibit larger sensitivity.
- This is supportive evidence that the prepayment-driven demand channel is not only a pricing phenomenon but also a *portfolio allocation* phenomenon.

6 Integrated takeaway: one coherent mechanism across prices, issuance, and institutions

Putting everything together, my coherent reading is:

1. Agency MBS exhibit a large convenience premium relative to AAA corporates and a smaller (in fact negative spread) relative to Treasuries, which is consistent with $\lambda_t^M \in (0, 1)$.
2. Mortgage-rate-driven prepayment incentives are a central, empirically powerful shifter of λ_t^M :
 - higher mortgage rates $\Rightarrow$ lower convenience premium and lower issuance (Tables 2–Table 4);
 - complementary prepayment-uncertainty evidence appears in dealer dispersion (Table 6);
 - the same logic extends to seasoned MBS (Table 7).
3. Institutional forces that change the safety/regulatory status of FN MBS move the convenience premium in event windows, consistent with λ_t^M shifting:
 - conservatorship increases FN safety relative to GN (Table 5);
 - LCR haircuts reduce FN relative regulatory convenience (Table 5).
4. Investor holdings respond in the direction implied by the demand channel (Tables 8 – Table 9), suggesting the mechanism operates through equilibrium portfolios, not only through marginal pricing.

7 Conclusion (What the paper establishes)

7.1 Core message

Agency mortgage-backed securities (MBS) guaranteed by Fannie Mae, Freddie Mac, and Ginnie Mae behave like a major class of *long-term safe assets*. The paper’s central empirical message is that newly issued (“production-coupon”) agency MBS embed a sizable *convenience premium*—meaning investors accept a lower yield because these securities provide liquidity/safety services—and that this convenience component is quantitatively large relative to classic benchmarks.

7.2 Main empirical conclusions to remember

1. **Magnitude of the convenience premium.** Using yield spreads against long-term Treasury securities and AAA corporate bonds (with adjustments for prepayment option values, duration differences, and default risk), the paper finds that production-coupon agency MBS carry a meaningful convenience premium. In particular, agency MBS exhibit:
 - a *convenience premium relative to AAA corporate bonds* (AAA–FN30y spread typically positive), and
 - an *inconvenience discount relative to Treasury securities* (Treasury–FN30y spread typically negative),consistent with “MBS are safe and liquid, but not as safe/liquid as Treasuries.”

2. **Prepayment-driven demand channel.** The convenience premium and issuance of newly issued agency MBS vary *negatively* with the at-issuance mortgage rate. Intuitively, a higher mortgage rate makes future prepayment/refinancing outcomes harder to price and more uncertain (greater prepayment-related complexity/uncertainty), reducing the desirability of MBS as safe assets, which lowers their convenience premium and issuance.
3. **Policy channels matter.** Two institutional forces materially shift the safe-asset demand for agency MBS:
 - **Government guarantee channel:** moving Fannie/Freddie into conservatorship in 2008 strengthens government backing and increases safe-asset demand (raising MBS convenience value relative to a control group).
 - **Regulatory treatment channel:** liquidity regulation (Basel III LCR) changes how banks value agency MBS as HQLA through haircuts, shifting demand and hence convenience premia.
4. **Broader implication emphasized by the paper.** Treating agency MBS as safe assets provides a unifying lens for housing finance, monetary policy transmission, and asset pricing: the convenience-premium component can affect funding costs and the response of MBS markets to macro and policy shocks.

8 Intuition (Why these results are economically natural)

8.1 Safe-asset pricing: a liquidity-services view

A clean way to organize the logic is: *safe assets deliver utility/transactions services in addition to their cash flows*. When an asset provides these services, investors accept a *lower* financial return.

A stylized representation used by the paper is a money-in-the-utility style formulation:

$$C_t \equiv c_t + \gamma_t v(Q_t), \quad (14)$$

$$Q_t \equiv Q_t^B + \lambda_t^M Q_t^M, \quad (15)$$

where:

- Q_t^B is (real) Treasury holdings, Q_t^M is (real) agency MBS holdings,
- $v(\cdot)$ is increasing and concave (liquidity services have diminishing marginal utility),
- γ_t shifts *overall* demand for safe assets (flight-to-safety, liquidity demand, etc.),
- λ_t^M captures *relative* “moneyiness” or safe-asset desirability of agency MBS compared with Treasuries.

From the Euler-equation logic for one-period yields, the paper highlights a convenient interpretation:

$$r_t - r_t^M = \lambda_t^M \gamma_t v'(Q_t) \quad (\text{MBS convenience premium}), \quad (16)$$

$$r_t - r_t^B = \gamma_t v'(Q_t) \quad (\text{Treasury convenience premium}), \quad (17)$$

$$r_t^M - r_t^B = (1 - \lambda_t^M) \gamma_t v'(Q_t) \quad (\text{MBS inconvenience relative to Treasuries}). \quad (18)$$

Key economic takeaway. If $\lambda_t^M < 1$, then agency MBS provide convenience, but less than Treasuries. This generates the empirical pattern “MBS look special relative to corporates, yet still trade cheaper than Treasuries.”

8.2 Why mortgage rates matter: the prepayment-driven demand channel

Agency MBS are *prepayable*: mortgage borrowers can refinance and prepay, making the timing of principal cash flows state-dependent. The paper’s prepayment channel can be summarized as:

$$\lambda_t^M = -(\text{prepayment model complexity}_t) \times f(m_t), \quad f(\cdot) > 0, \quad (19)$$

where m_t is the at-issuance mortgage rate for newly issued MBS.

A convenient reduced-form description of prepayment intensity is:

$$p_s^t = x_s + y_s \max\{0, m_t - m_s\}, \quad s \geq t, \quad (20)$$

so refinancing incentives (moneyness) depend on future mortgage rates m_s relative to the original rate m_t .

Intuition in words (connecting the model to the data).

- When m_t is higher, the security is more exposed to scenarios in which future rates fall meaningfully, making refinancing/prepayment waves more likely and harder to forecast.
- More uncertainty/complexity around future prepayment lowers the “safe-asset” desirability of newly issued MBS (a decline in λ_t^M).
- A lower λ_t^M reduces the MBS convenience premium and reduces issuance in equilibrium (because the securitization sector finds it less profitable to supply MBS when the convenience wedge is smaller).

8.3 Why policy events move convenience premia

Two institutional mechanisms change λ_t^M (relative safe-asset desirability of agency MBS):

- (i) **Government guarantee channel.** When government backing becomes stronger and more explicit (e.g., conservatorship of GSEs), the perceived “safety” of agency MBS principal increases, pushing up demand and convenience value.
- (ii) **Regulatory treatment channel (HQLA haircuts).** Liquidity regulation can mechanically shift bank demand by changing how much HQLA value a given dollar of MBS delivers after haircuts. A higher haircut reduces the effective liquidity service flow, lowering demand and convenience value.

8.4 A final “big picture” intuition

Put together, the paper’s conclusion is consistent with a simple hierarchy:

Treasuries (highest convenience) $\succ$

Agency MBS (meaningful but lower convenience) $\succ$ AAA corporates (lowest convenience among these). Agency MBS earn their safe-asset status from government-related protections and market/regulatory usage (HQLA, repo collateral), but they are still distinct from Treasuries due to prepayment-related cash-flow uncertainty and the associated pricing complexity.

9 Short concluding paragraph

The main lesson is that thinking of agency MBS as long-term safe assets is not just a label: it organizes their pricing (convenience premia), their issuance dynamics (endogenous supply responding to convenience value), and their sensitivity to mortgage rates and policy/regulation. This lens suggests that housing-finance outcomes and monetary-policy transmission can operate partly through fluctuations in the convenience value of agency MBS, not only through traditional interest-rate channels.

References

- [1] He, Zhiguo, and Zhaogang Song (2026). Agency MBS as Safe Assets. *The Review of Financial Studies*, 39(2), 387–426.